

LITERATURE STATUS NOTE FOR ARXIV:1610.05199

AGENT_LANE_10

STATUS

Status: literature_already_answered. The open-question signal in van Handel’s *Chaining, Interpolation, and Convergence II: The contraction principle* (arXiv:1610.05199) points to the discussion after Corollary 3.3 and to Section 3.3. There the author asks whether the positive-semidefinite assumption in the random-matrix chaining bounds can be weakened, and notes that Latala’s conjecture for symmetric Gaussian matrices with independent, non-identically distributed entries would follow from such an extension.

SOURCE QUESTION

In arXiv:1610.05199, after Corollary 3.3, the paper observes that the positive-semidefinite hypothesis ensures that the auxiliary functional $\|\cdot\|$ is a norm and that $\|v\|_z \leq \|v\| \|z\|$. It then asks whether this positive-semidefinite assumption can be weakened. Section 3.3 makes the consequence explicit: for a symmetric Gaussian matrix with independent entries $X_{ij} \sim N(0, b_{ij}^2)$, Latala’s conjecture predicts

$$\mathbf{E}\|X\| \asymp \mathbf{E} \max_i \left(\sum_j X_{ij}^2 \right)^{1/2}.$$

ANSWERING SOURCE

The primary answering paper is Latala–van Handel–Youssef, *The dimension-free structure of nonhomogeneous random matrices* (arXiv:1711.00807; Invent. Math. 214 (2018), 1031–1080). Its abstract states that the main result settles, for $p = \infty$, a conjecture of the first author. Theorem 1.1 proves that if $X_{ij} = b_{ij} g_{ij}$ is a symmetric matrix with independent standard Gaussian variables g_{ij} for $i \geq j$, then

$$\mathbf{E}\|X\|_{S_\infty} \asymp \max_i \left(\sum_j b_{ij}^2 \right)^{1/2} + \max_{ij} b_{ij}^* \sqrt{\log i},$$

with universal constants. The preceding paper arXiv:1502.05003 identifies this expression with the row-norm formulation of Latala’s conjecture, and a later contextual source, arXiv:2112.14413, explicitly records that the Latala–van Handel–Youssef result finally answered the conjecture in the affirmative.

SCOPE AND CAVEAT

This is an already-known literature answer, not a new proof produced in the run. The packet records that the main open consequence highlighted in arXiv:1610.05199, namely Latala’s Gaussian spectral-norm conjecture, was settled by arXiv:1711.00807. It does not assert that the particular positive-semidefinite chaining corollary from arXiv:1610.05199 was itself extended verbatim.

SEARCH NOTES

Cheap run indexes were searched for arXiv:1610.05199, contraction principle, Latala, Gaussian random matrices, positive semidefinite assumption, and the label **supernck**. No existing solution, attempt, proof-gap, or registry entry for this source question was found. Local source search found arXiv:1711.00807 as the primary answer and arXiv:2112.14413 as independent later confirmation of the answer status.